

ESMValTool

Earth System Model Evaluation Tool

An Introduction to ESMValTool – Advanced Topics

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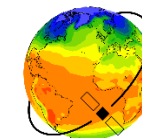
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ESMValTool Recipes

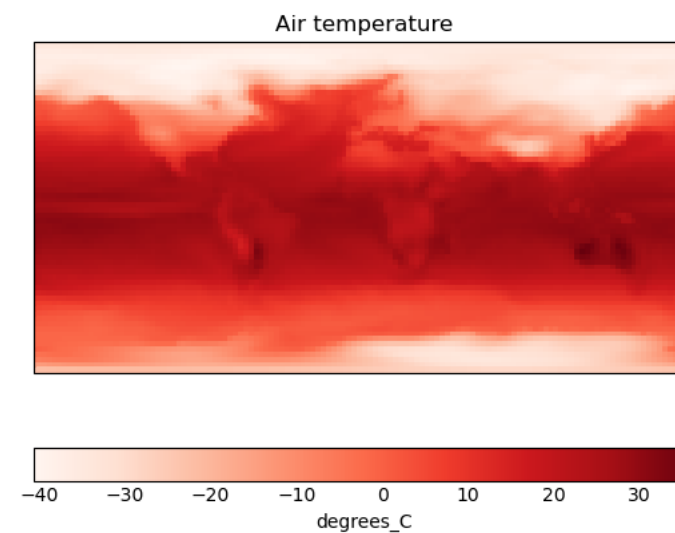
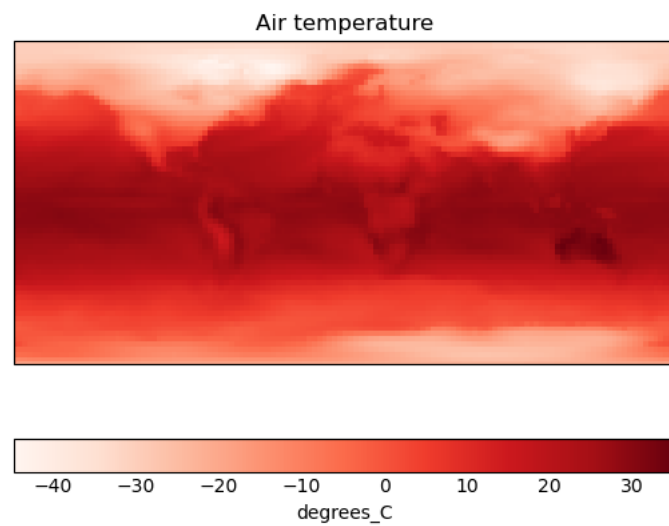
ESMValTool is controlled through a **recipe** (text file in YAML format).

Recipes contain

- Datasets to analyze
- Preprocessors to use
- Variables to analyze
- Diagnostic scripts to use (Python, NCL or R scripts)



Diagnostic Output



Running ESMValTool

- Before running ESMValTool:
 - load module or activate conda environment.
 - Ensure config files are in place
 - Run example recipe as per Tutorial lesson on writing own diagnostic.
- If *run* is successful, in the `esmvaltool_output` directory, files and subdirectories will be created under `recipe_python_#_#`:
 - `index.html`
 - `run`
 - `preproc`
 - `plots`
 - `Work`
- This directory structure has files (preprocessor output, data specifications etc.) for diagnostic Python script to read.

Where does the output go?

For preprocessed output:

output directory -> preproc -> map (diagnostic name)
->tas (variable name)

which has:

- processed files (netCDF)
- provenance files – information about the data and how it was generated (xml files)
- **metadata.yml** file – an entry for every variable in every diagnostic – interface file for preprocessor output.

Where does the output go?

For diagnostic output:

output directory -> work/plots/ -> map (diagnostic name)
->script1 (script name in recipe file)

which have:

- processed files (netCDF) or plots
- provenance files – information about output and how it was generated (xml files)
- citation information

Where does the output go?

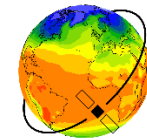
For diagnostic output:

output directory -> run ->

which has:

- log files, resource usage information
- map (diagnostic)
 - provenance file
 - diagnostic specific log
 - resource usage
 - **settings.yml** - interface file

Helpful utility functions



```
8  
9     from esmvaltool.diag_scripts.shared import (  
10         group_metadata,  
11         run_diagnostic,  
12         save_data,  
13         save_figure,  
14         select_metadata,  
15         sorted_metadata,  
16     )  
17     from esmvaltool.diag_scripts.shared.plot import quickplot  
18
```

Questions?

https://docs.esmvaltool.org/en/latest/introduction.html#support

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Thanks!

